

Instructions:

Data → Data Analysis → Regression.

Y Range: Total_Weekly_Sales (with header) | X Range: Store_Size (with header)

Check: Labels, Residuals, Residual Plots, Line Fit Plots | Confidence: 95%.

Output the full regression table starting at A7.

Q10a R-squared (interpret)	Q10b Significance F ($\alpha=0.05$)
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R-squared is .691666722, which means 69.71% of the variation in total weekly sales is explained by Store Size

Yes, the model is statistically significant. Significance F is very close to zero.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.83166503
R Square	0.691666722
Adjusted R Square	0.690946317
Standard Error	367654.776
Observations	430
ANOVA	
	df
Regression	1
Residual	428
Total	429
Coefficients	
Intercept	18571.9738
Store_Size	8.856472917

RESIDUAL OUTPUT

Observation	Predicted Total_Weekly_Sales
1	1358689.173
2	1358689.173
3	1358689.173
4	1358689.173
5	1358689.173
6	1358689.173

7	1358689.173
8	1358689.173
9	1358689.173
10	1358689.173
11	1358689.173
12	1358689.173
13	1358689.173
14	1358689.173
15	1358689.173
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17	1358689.173
18	1358689.173
19	1358689.173
20	1358689.173
21	1358689.173
22	1358689.173
23	1358689.173
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27	1358689.173
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32	1358689.173
33	1358689.173
34	1358689.173
35	1358689.173
36	1358689.173
37	1358689.173
38	1358689.173
39	1358689.173
40	1358689.173
41	1358689.173
42	1358689.173
43	1358689.173
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45	1810298.44
46	1810298.44
47	1810298.44
48	1810298.44
49	1810298.44
50	1810298.44

51	1810298.44
52	1810298.44
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54	1810298.44
55	1810298.44
56	1810298.44
57	1810298.44
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59	1810298.44
60	1810298.44
61	1810298.44
62	1810298.44
63	1810298.44
64	1810298.44
65	1810298.44
66	1810298.44
67	1810298.44
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69	1810298.44
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84	1810298.44
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88	349733.2091
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135	1841792.058
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418	1139022.076
419	1139022.076
420	1139022.076
421	1139022.076
422	1139022.076
423	1139022.076
424	1139022.076
425	1139022.076
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427	1139022.076
428	1139022.076
429	1139022.076
430	1139022.076

der)

Q10c Coef Store_Size (interpret)	Q10d p-value of Store_Size (conclude)
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The coefficient for Store Size is 8.856472917, which means for every additional square foot of store size, weekly sales increase by 8.86. If a store was increased by 1000 sq ft, sales would be expected to go up by 8,860. Yes, Store Size is a significant predictor of sales. It's P-value is extremely close to zero, indicating high statistical significance.

<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
1.29778E+14	1.29778E+14	960.1084	2.1E-111
5.78528E+13	1.3517E+11		
1.87631E+14			

<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
41490.51107	0.447619789	0.654654	-62978.5	100122.5
0.285825304	30.98561534	2.1E-111	8.294677	9.418269

<i>Residuals</i>
282356.0967
100911.9967
35704.6667
-39363.5833
277650.4767
443788.2567

461180.8267
180698.6567
329731.5867
316741.9867
318783.6067
152378.8967
290915.4567
540987.7067
262342.5267
162888.6967
110239.1967
325830.8167
252406.8767
237212.6967
196755.3767
338541.7867
271917.8267
169156.6367
181732.3167
411164.9867
168324.8667
139265.5867
80434.5367
272446.6167
233720.7967
239178.8767
135433.2067
223394.2267
303078.1567
158739.6967
147436.8867
78370.0867
312096.7967
214383.6367
149379.5967
134970.5667
175113.0567
-10778.30031
-65572.96031
-98529.33031
-149392.3003
125001.4997
293024.2397
386390.0197

51504.25969
142257.2197
127329.8197
165783.6897
-19859.28031
47182.39969
318737.4697
125570.6597
37046.00969
-46165.35031
113658.6497
107222.5497
190642.2297
102492.6497
99793.92969
199918.0497
152625.8597
77434.76969
70747.67969
231208.9597
19776.68969
9368.019691
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135806.1997
56421.51969
117717.5697
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4508.189691
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90446.68969
37691.96969
24159.90969
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16085.40087
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75227.45087
123559.2609
125857.8709
69192.26087

120019.3509
95428.84087
62042.59087
64174.04087
57755.63087
153498.9209
71056.53087
85088.92087
44882.90087
90180.36087
82252.15087
68379.55087
63968.08087
82535.32087
96603.59087
92341.58087
69764.74087
73232.12087
61473.29087
67179.89087
82691.64087
39694.69087
70257.08087
42078.39087
45185.62087
62716.46087
59105.52087
58496.52087
57855.95087
64658.88087
40079.81087
93824.44087
61071.18087
74779.87087
55699.49087
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99884.552
163305.702
86928.452
331581.852
532868.582
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364527.842

360658.752
373175.382
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247589.712
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365422.752
312345.612
337568.882
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-234612.2119
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160802.8668
49311.14679
41505.94679
41525.65679
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628449.4045
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672540.8045
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568459.8245
666977.7145
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730944.9545
678581.5845
800418.0145
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625962.0745
511263.4645
569261.2045
501146.9145
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555840.3345
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574867.0345
595812.7445
605326.9745

lower 95.0%		upper 95.0%	
-62978.5	100122.5		
8.294677	9.418269		



